



دانشگاه صنعتی امیر کبیر
(پلی تکنیک تهران)

Computer Architecture

Chapter 1 Introduction

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عظیم فرقدان 

بهمن ۱۴۰۳

- 1. About the Instructors**
- 2. Assessment**
- 3. Reference Books**
- 4. Computers**
- 5. Computer History**
- 6. Moore's Law**
- 7. Computer Architecture**
- 8. Conclusion**

- **Azim Farghadan**

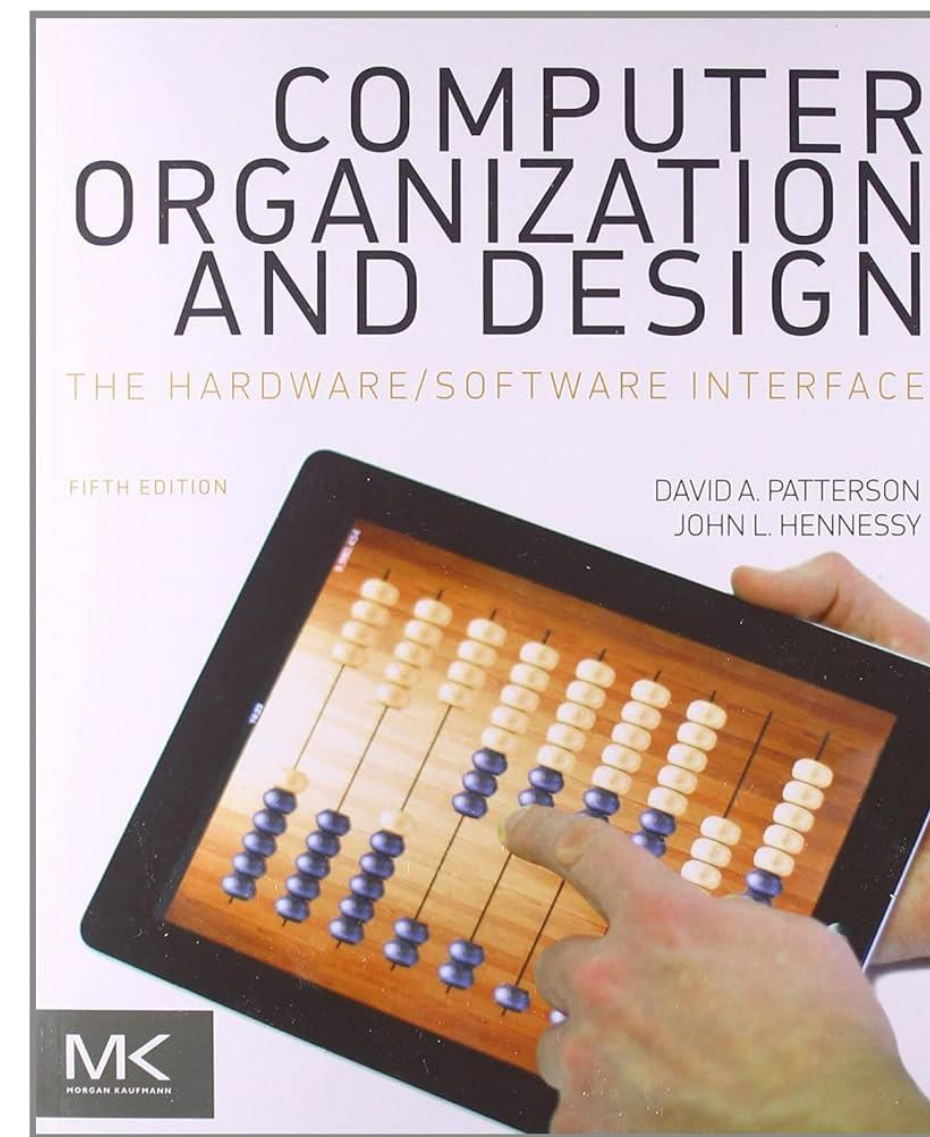
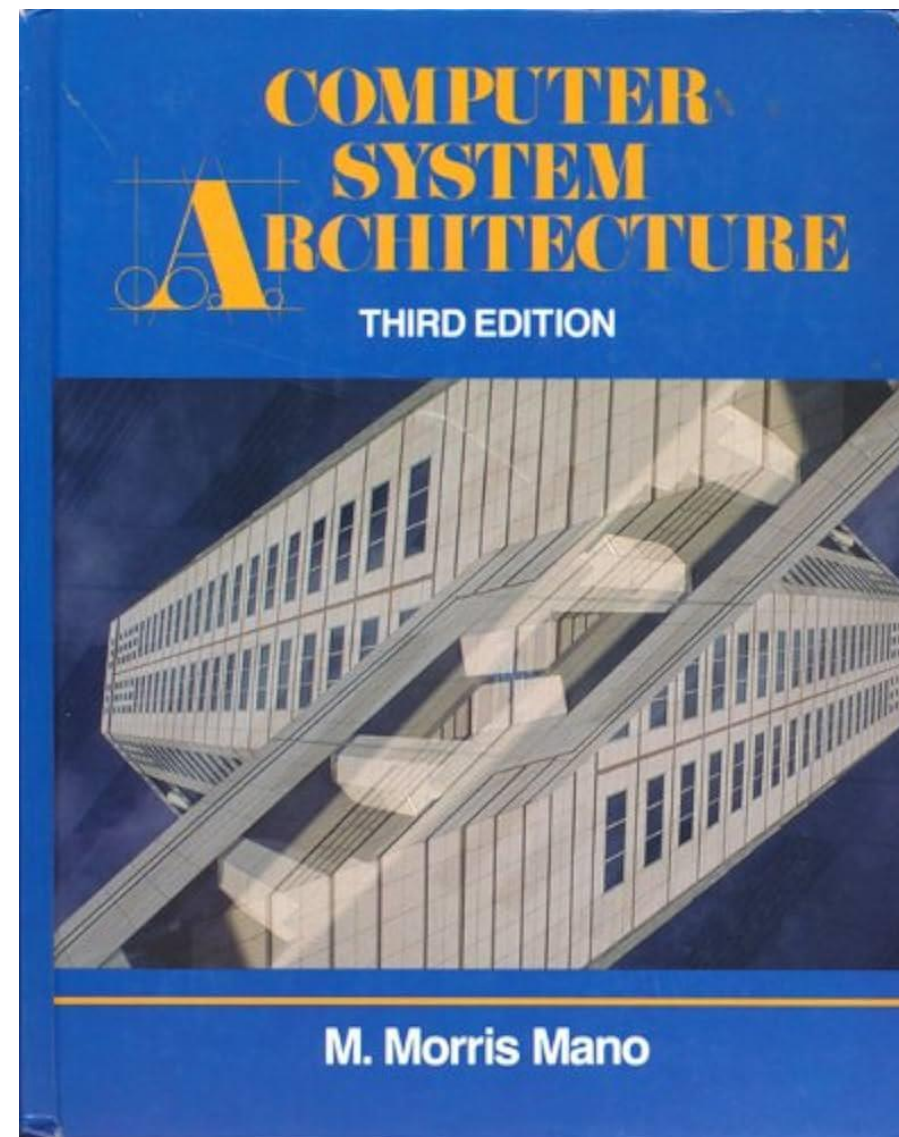
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- **Teaching Assistance Team:**

- TA telegram channel: t.me/CA_1404
- Head TA: Helia Akbari
- TA team: Sina Farahani, Alireza Yousefpour, Sina Negahban, Parand Khalili

Method	Quantity	(%)	(%)
Quiz	-	5	-
Homework/Problem Solving	7	15	-
Laboratory	-	-	100
Midterm Exam	1	35	-
Final Exam	1	45	-
Attendance & participation	-	extra	-
Final Grade		75	25

1. **M. Morris Mano, “Computer System Architecture”, Third Edition**
2. David A Patterson and John L. Hennessy, “Computer Organization and Design”, Fifth Edition

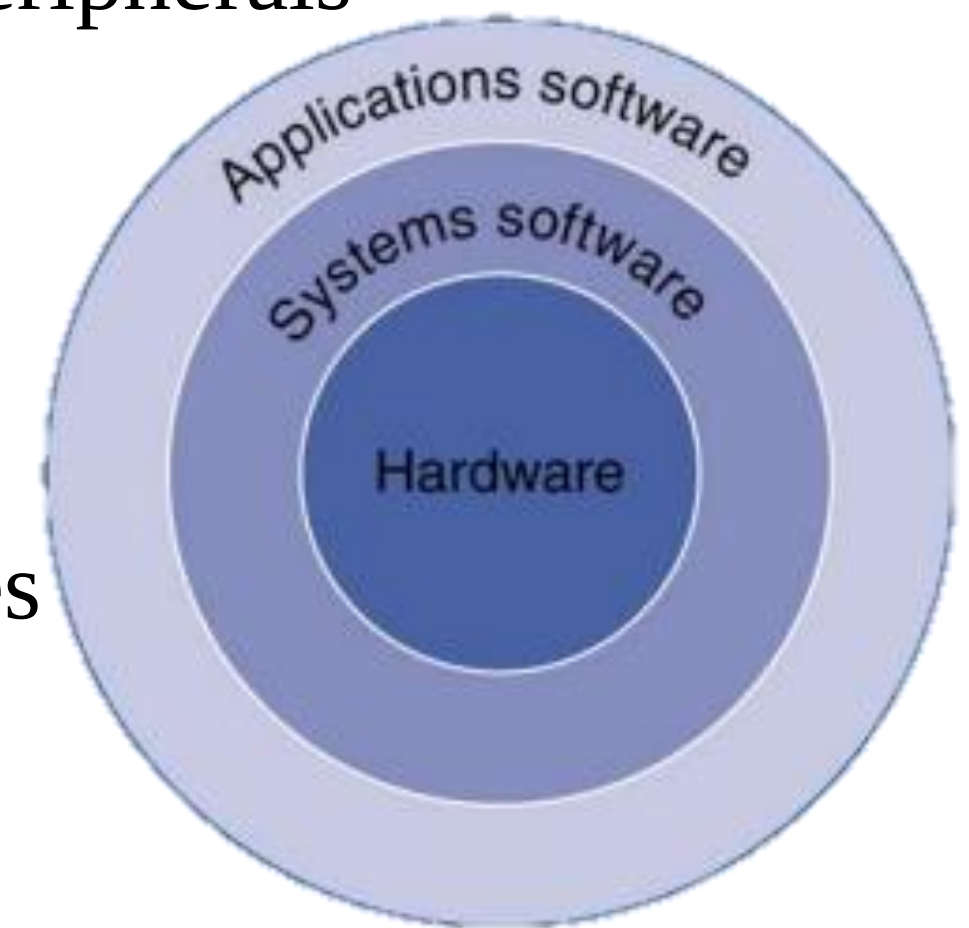


What is a Computer?

- An **electronic device** capable of **storing and processing information** in accordance with a predetermined set of **instructions**
- A **machine** that manipulates **data** according to a set of **Instructions**
- A **general purpose** device that can be **programmed** to carry out a **set of arithmetic or logical operations**
- Examples:



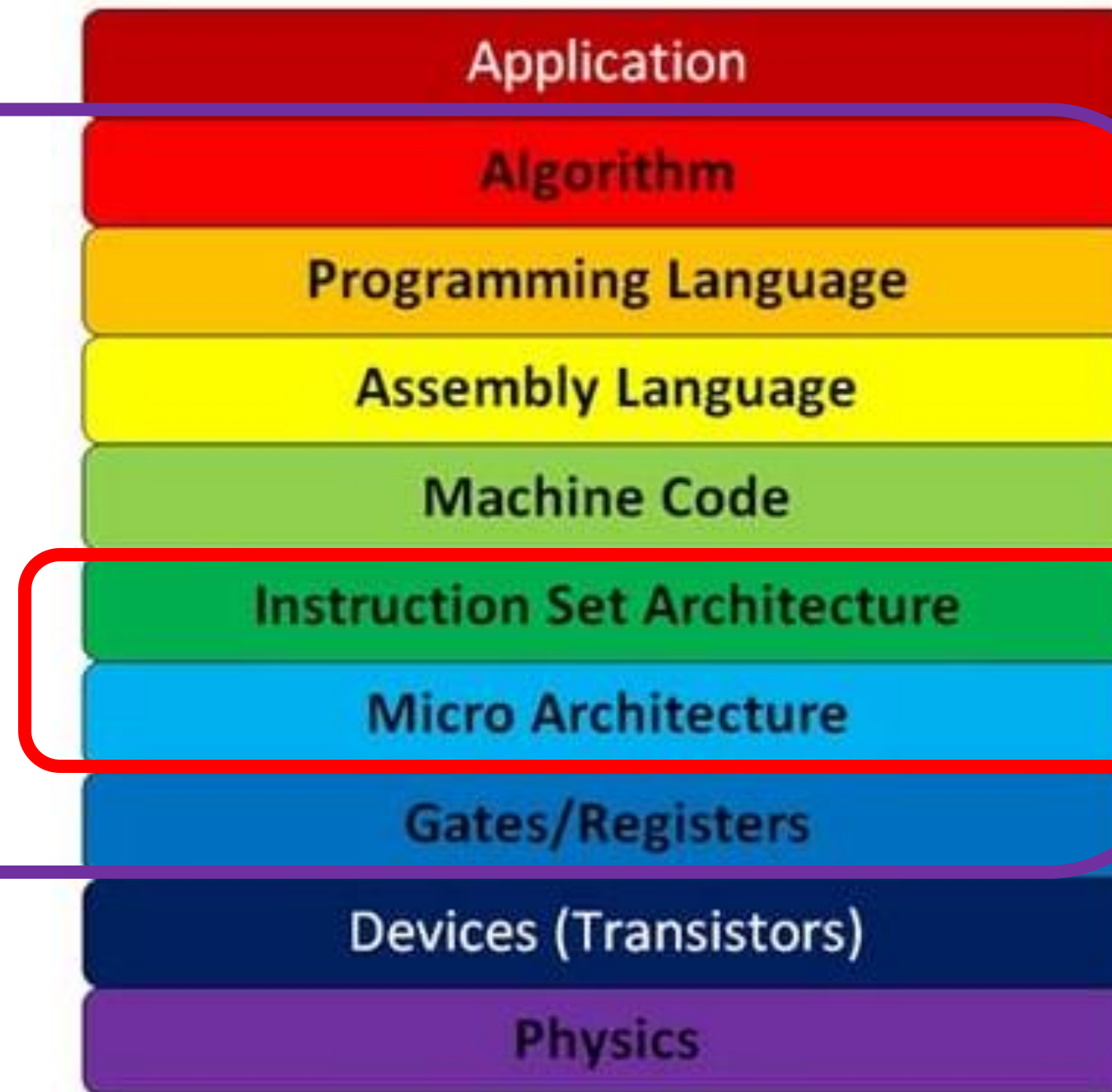
- A computer system consists of **hardware** and **software** that are combined to provide a tool to solve problems (**with best performance**)
- Hardware may include:
 - CPU, memory, disks, printers, screen, keyboard, mouse, Other peripherals
- Software may include:
 - System software: Compiler, Operating System
 - Application software: Programs written with high-level languages



Hierarchical Abstraction Levels

- We can comprehend complex computer systems by hierarchical abstraction levels

Computer Architecture
(broad view)



Software

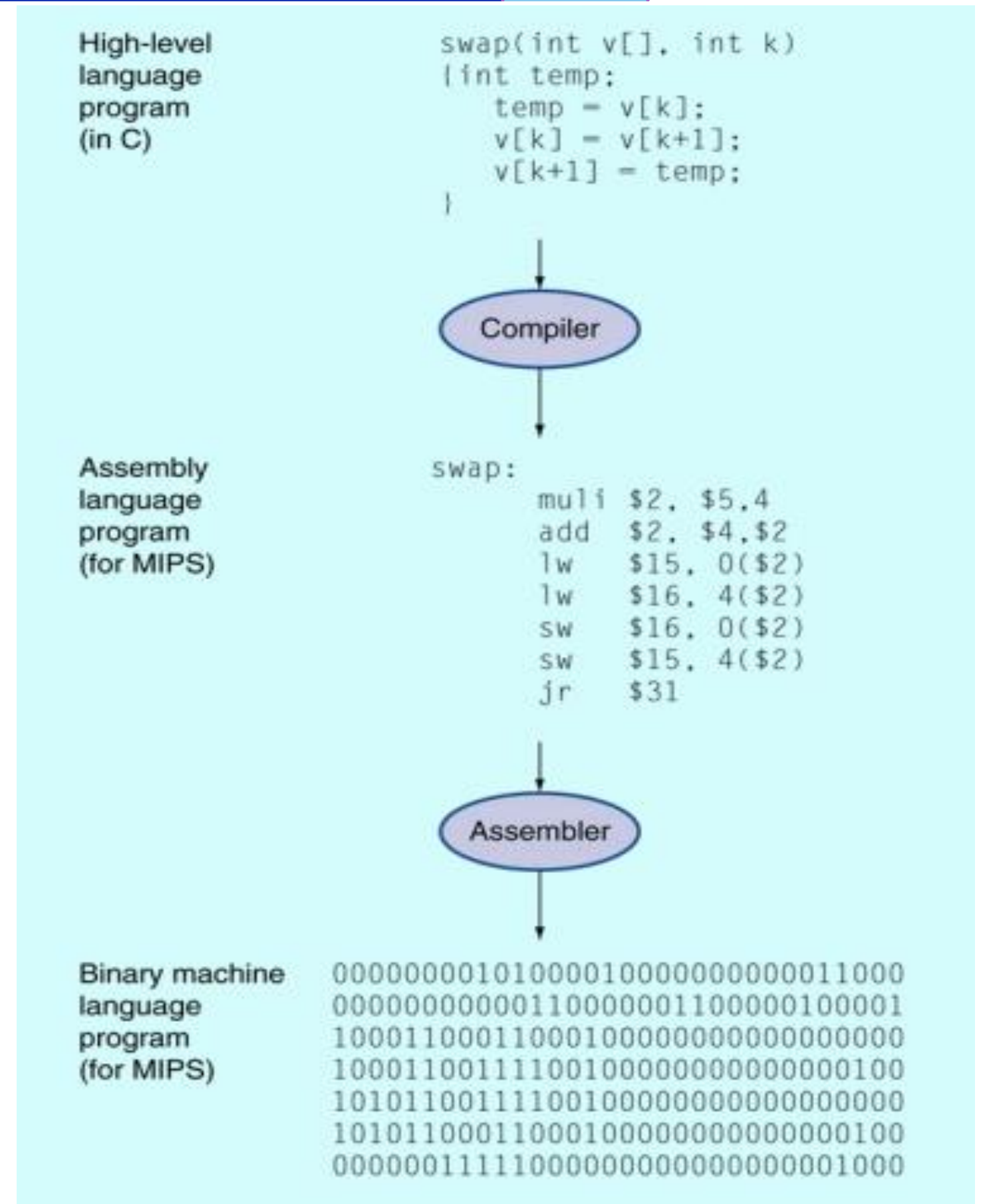
Hardware

Computer Architecture
(narrow view)

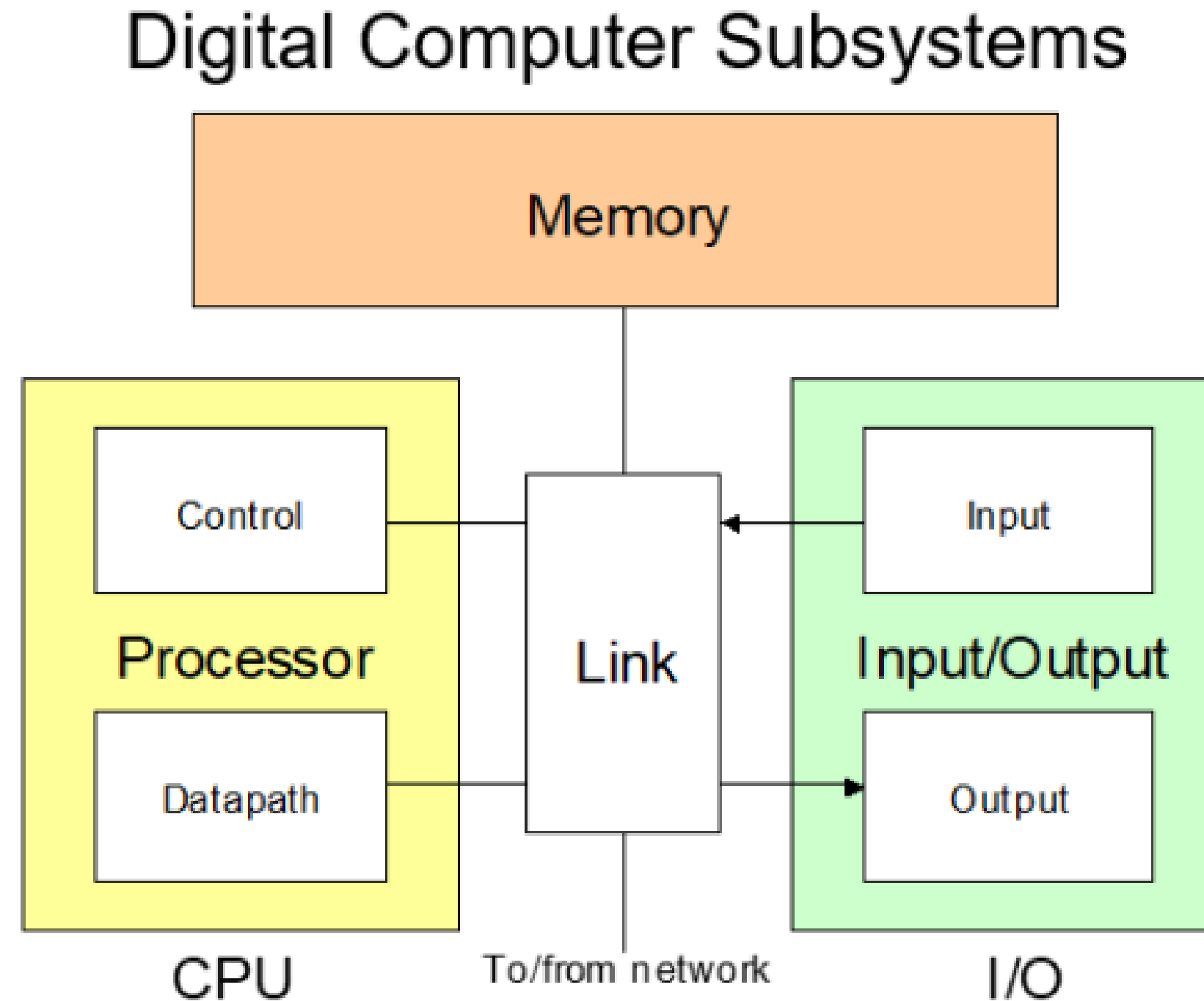
Layers of Abstraction

Programming Languages Levels

- High-level languages
 - Closer to natural language
 - portability and performance
- Assembly language
 - Symbols to be replaced with machine code
- Machine language
 - A string of 0s and 1s



- Input
- Output
- Memory
- Control unit
- Data path



Types of Computer Systems

- Desktops
- Servers
- Embedded Systems

■ Desktops

- General purpose computers
- Variety of software
- Designed for use by individuals
 - PCs, Notebooks



■ Servers

- High-end computing systems
 - Consists of several CPUs
- Used by multiple users
- Network based
- High capacity, performance, reliability
- Range from small servers to building sized



■ Embedded Systems

- A computer system designed to perform one or a few **dedicated** functions
- Often with **real-time, power, high reliability, low-cost, size/weight** constraints
- **Hidden** as components of systems



- Mechanical Computers

- Abacus



- Mechanical calculators

- Pascal's calculator (1642)



- Curta (1930s)



- Punched-card data processing (1725, 1800s, 1900s)



- Electro-Mechanical Computers

- Switches, relays

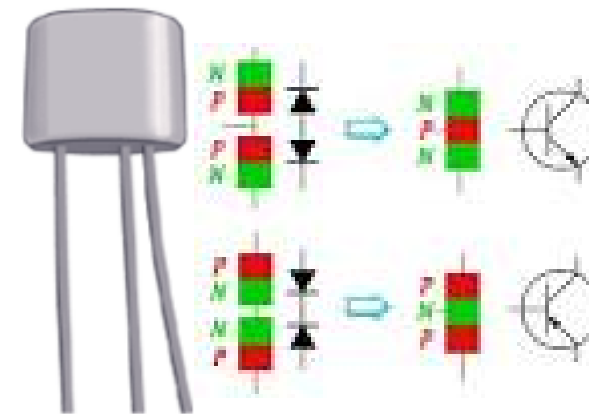
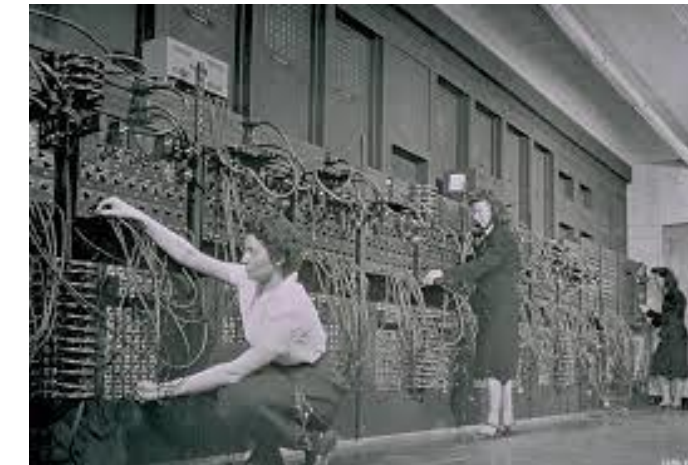
- Early 1900s

- Zuse's Z2, Z3, Z4 (world's first commercial digital computer, mechanical memory)



■ Electronic Computers

- 1st generation
 - Vacuum tubes (1945-1955)
 - ENIAC, 1946
- 2nd generation
 - BJT transistors
 - CDC 6600 (1946-1969)
 - World's fastest computer
 - \$2,370,000
 - 30kW, 6tons
 - 10MHz, 2 MIPS
 - Up to 982 kilobytes



- Electronic Computers

- 3rd generation

- integrated circuits (1965-1974)

- Intel 4004 (1971)

- The first commercially available microprocessor

- 10 μm process

- 92,000 instructions per second

- 1 MHz clock rate

- 4th generation: VLSI

- 1974-1989

- 5th generation: ULSI

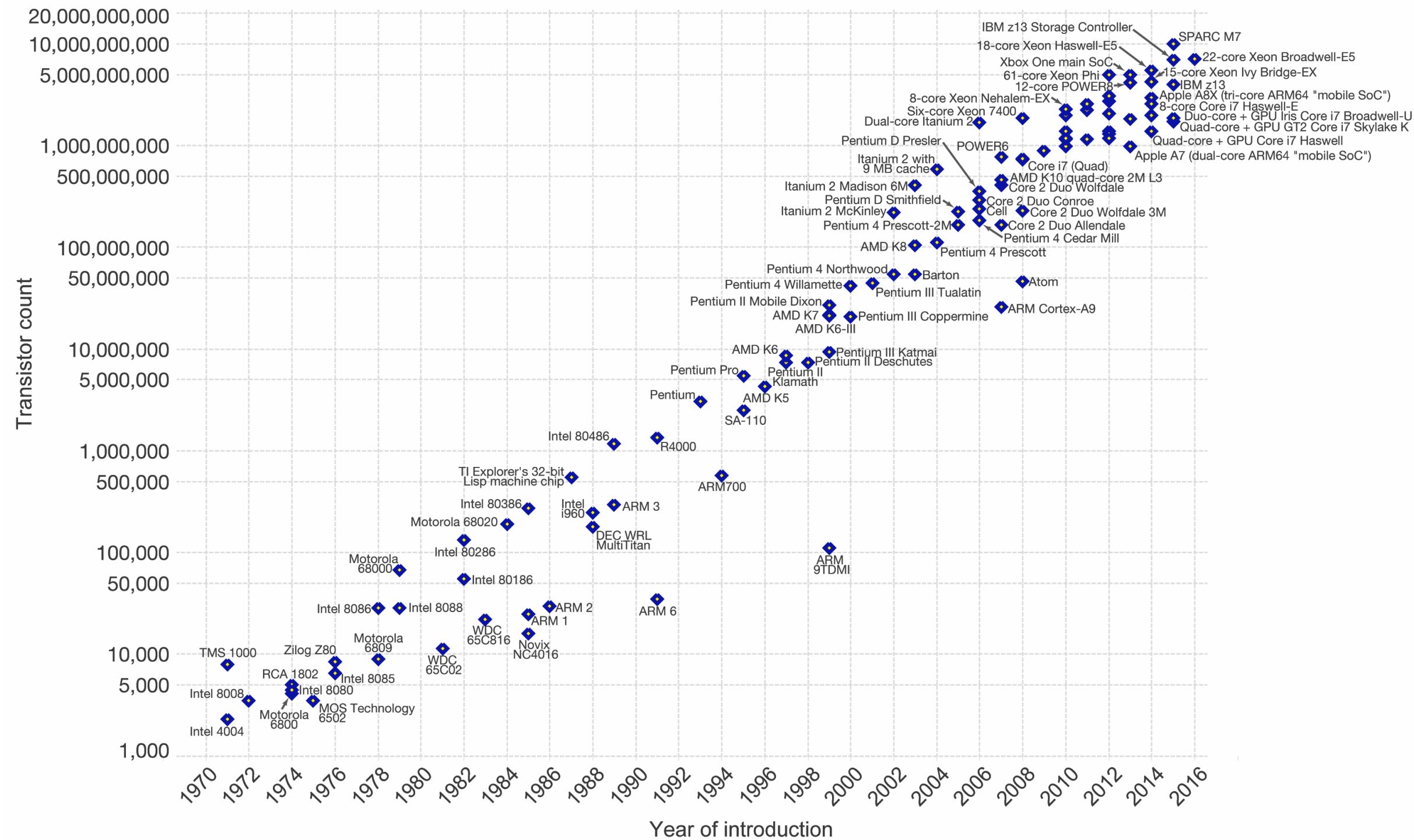
- 1990-present



Year	Technology Used in Computers	Relative Performance per unit cost
~1950	Vacuum tube	1
1960s	Transistor	35
1975	Integrated circuit	900
1995	Very large scale integrated circuit	2,400,000
2005	Ultra large scale integrated circuit	6,200,000,000

Moore's Law

- from 1970 Transistor count doubles every 24-month



- Knowledge and art of designing, choosing, and connecting hardware components, designing hardware/software interface for computational systems in order to reach certain efficiency goals.
- Definition of efficiency is dependent on different platforms and usecase:
 - Reliable, secure, and safe
 - Memory-centric
 - Low latency
 - High performance

Conclusion

- ✓ A modern computer is an electronic, digital, general purpose computing machine that automatically follows a step-by-step list of instructions to solve a problem. This step-by step list of instructions that a computer follows is also called an algorithm or a computer program.
- ✓ Why study computer organization and architecture?
 - ✓ Design better programs, including system software such as compilers, operating systems, and device drivers.
 - ✓ Optimize program behaviour.
 - ✓ Evaluate (benchmark) computer system performance.
 - ✓ Understand time, space, and price trade-offs. Computer organization
 - ✓ Encompasses all physical aspects of computer systems.
 - ✓ E.g., circuit design, control signals, memory types.
 - ✓ How does a computer work?



Thanks
